Project Peer Review Feedback - BMIMU Video

January 2026

1 Strengths of the paper

This paper demonstrates several notable strengths that made it very clear and easy to understand for someone who was not part of the project.

First of all, the project is very well structured, and the order of sections helps the reader to understand everything at the right time. The background section effectively explains all complex concepts, which is very useful to understand the project later on in the report.
Terms such as IMU sensors, pose estimation models (PoseNet vs. MoveNet), and event-based synchronization were very clearly explained.

In addition, the authors make good use of figures, especially the workflow diagrams in Figures 2, 3, and 4, which clearly illustrate the concepts and allow the reader to have some kind of clear map in their head to understand the structure of the import/export logic.

Furthermore, the appendices offered several pictures of the website, which allow the reader to really see and understand how the interface works. Appendix B provides a comprehensive user guide that shows all the steps as if I were to use the website myself, which I believe is very useful to see everything that was previously explained in the report put into practice.

The paper is generally well-presented and demonstrates a clear organizational structure, and easy to understand language using clar terms and explanations. The logical flow from Introduction through Background, Methods, Results, Discussion, and Conclusion allows readers to follow the development and implementation of the system systematically.
Visual elements such as flowcharts, screenshots, and interface diagrams are well integrated and appropriately captioned, supporting the text rather than overwhelming it.

I believe that the report makes perfect use of the figures, as previously mentioned, especially in Appendix B where you could see exactly how the website works, the steps on how to use it, with clear images that allow the reader to have an interactive experience even with just a report, and not the website itself.

I believe that the report makes perfect use of the figures provided, as previously mentioned, especially in Appendix B where you can see exactly how the website works, the steps on how to use it, with clear images that allow the reader to have an interactive experience even with just a report, and not the website itself. The workflow diagrams in Figures 2, 3, and 4 also effectively illustrate the export/import processes in a visual way that complements the text

well. However, the report lacks tables entirely, which would have been useful for presenting quantitative data and technical specifications. Additionally, while one formula is provided for synchronization ($\Delta t = t_{Video} - t_{IMU}$), there are no data tables or results charts showing performance metrics. These absences represent notable gaps that will be discussed in more detail in the considerations section below.

2 Considerations

While the report successfully describes the system's features and functionality, it falls short in demonstrating how well the system actually performs.
The Results section is notably weak, primarily describing what the system can do rather than providing evidence of how effectively it does it. There is a critical absence of quantitative data throughout the report—no measurements of frame rates achieved, no accuracy metrics for pose detection, no validation of synchronization precision.
For instance, the authors claim that synchronization is successful but provide no testing to verify this; How do we know that this is in fact working. An experiment marking the same event multiple times and measuring the variance in calculated offset could have been useful to demonstrate their solution and in fact proove that it works.

In addition, the report also lacks references and literature review:
In the introduction, the authors state that commercial motion capture systems are expensive and require specialized equipment, yet they provide no citations to support these claims. In the motivation paragraph of the introduction, they mention how objective movement measurement is especially relevant in healhtcare and other applications but do not provide resources to literature or examples of real life applications of it. This would emphasize the need to develop such an easy to access tool. The use of only one reference is too little, and weakenes the research.

Similarly, technical decisions appear arbitrary and unjustified, such as the choice of 104 Hz sampling rate for the IMU.
The absence of rigorous testing protocols, comparison with ground truth data, or comparing it to existing solutions represents a significant gap in validating whether the system achieves its stated goals.

The authors do acknowledge many of these limitations in their Discussion section, openly stating that the system lacks numerical validation and that "functional validation" is not equivalent to "quantitative correctness."

3 Evaluation of Approach and Conclusion

The paper addresses the problem of expensive and complex motion capture systems by proposing a low-cost, browser-based alternative that combines IMU sensor data with video-based pose estimation. Their approach of integrating two complementary data sources is well justified. The decision to implement the system as a web application is practical and aligns well with their accessibility goals of making it accessible to everyone without any previous training.

The manual synchronization method, though functional, is a significant limitation that undermines the "low-complexity" claim, requiring users to manually identify and mark reference events is time-consuming and prone to human error, which contradicts the goal of making motion analysis accessible to non-experts.

To improve the work, the authors should implement rigorous testing protocols: validate synchronization accuracy through repeated trials, benchmark pose detection performance under various conditions (lighting, distance, occlusion), measure system performance metrics (frame rates, latency, processing time), and ideally conduct a small user study to assess usability.

Additionally, exploring automatic synchronization methods, such as detecting claps or jumps algorithmically from both video and IMU data, would significantly enhance the "low complexity" aspect of the system.

Finally, comparing their approach, even qualitatively, to existing low cost alternatives or open-source motion capture tools would better position their contribution within the broader field.